The social dynamics of data sharing

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Understanding the supply of demand for data

- The supply of data - data custodians
  - What are custodians' concerns
- The demand for data - secondary users
  - What do users expect?
- A model of data sharing
  - The Theory of Planned Behaviour
- Aligning supply and demand
  - Bridging the gap

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Why would researchers share data?

ADA's advice (https://ada.edu.au/depositing-data/):

- Your data will be stored in a **safe place**, where it is **easily accessible**, and in a form that can be understood
- The **potential impact** of your data is greatly increased. Other researchers, even from other disciplines, will be able to use your data (subject to the conditions you specify)
- Anyone who publishes results from your data is expected to **cite the data file** in their publications. Distribution of the data and documentation is managed without expense to the person who originally conducted the research.
- **Publicity** for your data – your data is published in the ADA data catalogue, raising the profile of your research.
- Sharing data through ADA allows you to **get in touch with other researchers with similar research data interests** to you.
- You can f**ind your own data in future** - you will have a preserved version of the data held with ADA.
- Your data will be given a **global persistent identifier** in the form of a DOI in the ADA Dataverse.
- **Versions of your data are stored and recorded** in the metadata.

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Why wouldn’t researchers share data?

Privacy and confidentiality:

- Is my data safe and secure?
- Can others identify my participants?

Intended use:

- Is my data being used “appropriately”?
- Can my data really be “understood” by others?

Intellectual property:

- Will someone else get there before me?
- Will I be “gazumped”?
- Will someone else find out what I did wrong?

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What do secondary users “need” (want)? (1)

“We emphasize that direct access to micro-data is critical for success. Alternatives such as access to synthetic data or submission of computer programs to agency employees will not address the key problem of restoring US leadership with cutting-edge policy-relevant research.”

Card, Chetty, Feldstein and Saez, 2010 (emphasis in original)

http://rajchetty.com/chettyfiles/NSFdataaccess.pdf

What do secondary users “need” (want)? (2)

Preferred: “Here's what you need to do if you want an anonymised 1% sample of the US Census

- Go to Google and type US Census 1% sample, click on link to the Census.
- Download each of the state files from the FTP site and merge them yourself. Or just check things out for one of the states. Whatever you like.
- Start mucking about to test whether your pet theory is plausible.

Non-preferred: Here's what you need to do if you want an anoymised sample of the NZ Census, or a Confidentialised Unit Record File (CURF) of any of big Stats series:

- Go to Stats NZ's site, here.
- Follow the instructions below: …”
- (Followed by several pages of instructions, Application Process, Assessment Criteria, Methods of Access, …)

Eric Crampton, the New Zealand Initiative, Wellington, formerly University of Canterbury.
http://offsettingbehaviour.blogspot.com.au/2015/10/curf-and-turf.html

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Can we bridge depositor and user expectations?
Remember: these are often THE SAME PEOPLE

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Understanding data sharing practices

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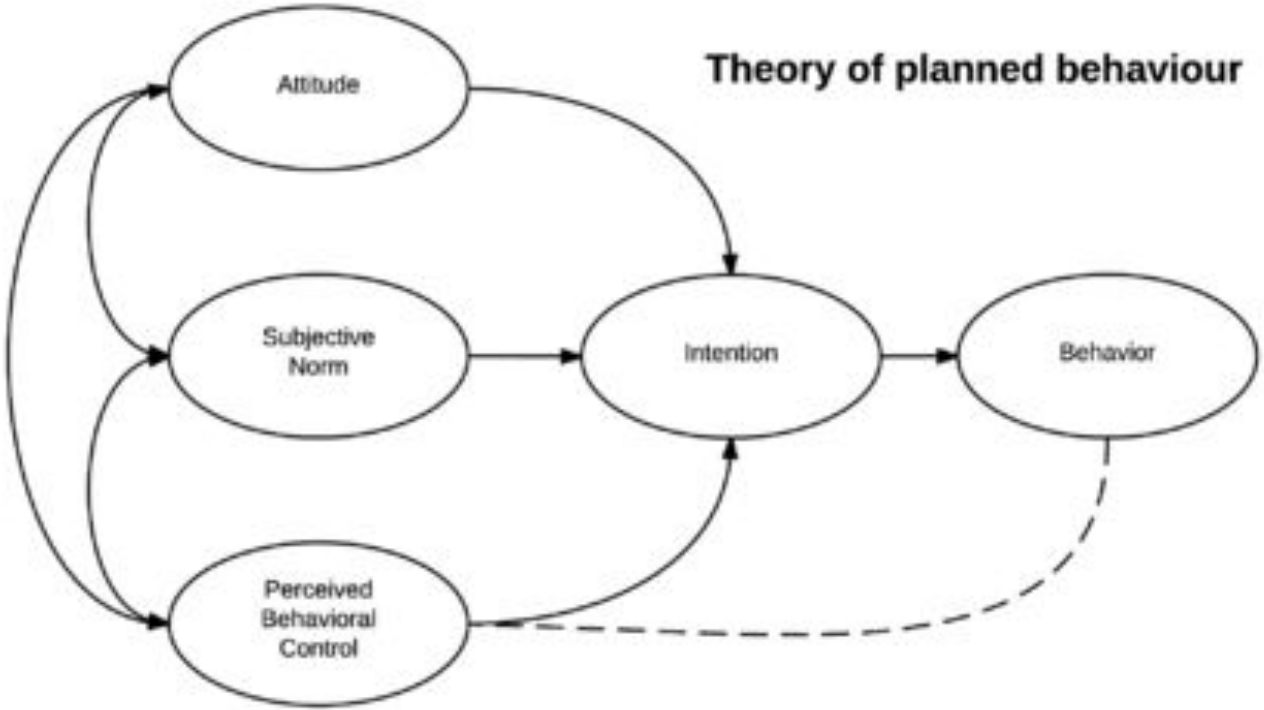
The Theory of Planned Behaviour



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The elements of the TPB

- Attitudes: attitudes towards the behaviour
  - E.g. I believe that data sharing is important
- Norms: perceptions of the judgment of significant others
  - E.g. My colleagues would expect me to share my data
- Perceived behavioural control: beliefs about control over one's ability to undertake the behaviour
  - E.g. I think it is relatively easy for me to share my data
- Behavioural intention: the intent to perform the action
  - E.g. I intend to share the data I produced in my research project
- Behaviour: the actual performing of the action
  - E.g. the deposit of the data in a repository

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McEachern and McDougall (2014)

Reanalysis of social science subset of Tenopir et al. (2011) - https://doi.org/10.1371/journal.pone.0021101

n=105

Measures:

- Core measures of TPB, plus
- Institutional practices
- Interest in data re-use (willingness ot use others' data)
- Previous data sharing

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Research questions (1)

1. What are the characteristics of those researchers with intention to perform data sharing behaviour?

- Positive attitude towards data sharing
- No effect of subjective norms among peers (although a weak measure used)
- LOW perceived behavioural control

- More likely to be female (but disappears if controls for data sharing and data reuse)
- No differences by age

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Research questions (2)

Are institutional data management procedures associated with intention to perform data sharing behaviour? NO – Not in this sample

Is past behaviour associated with intention to perform data sharing behaviour? YES

Is willingness to reuse data associated with intention to perform data sharing behaviour? YES – Strongest influence in model

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Kim and Stanton (2015)

https://doi.org/10.1002/asi.23424

Cross-discplinary survey of 1317 researchers

Distinguishes different elements of “individual characteristics”

- Career benefit
- Career risk
- Perceived effort
- Scholarly altruism

PLUS: Regulatory pressure from JOURNALS and FUNDERS

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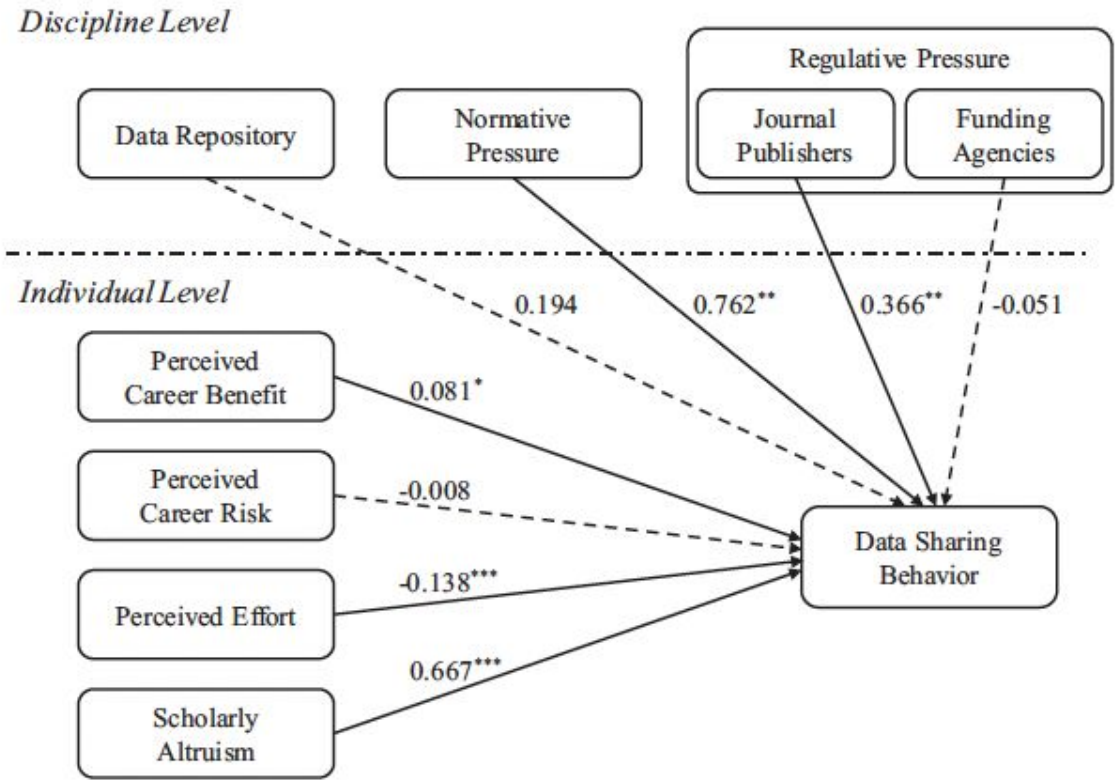

FIG. 2. Hypothesis testing results based on scientists' data-sharing behavior model. Unstandardized beta: ***$p < .001$; **$p < .01$; *$p < .05$.

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Zenk-Moltgen et al. (2018)

https://doi.org/10.1108/JD-09-2017-0126

- 426 political scientists and demographers
- “RQ4: there is a positive correlation between a **researcher’s attitude toward data sharing** and his/her behavioral intention to do so.
- RQ5: there is a positive correlation between **perceived social pressure** to share data with others and behavioral intention to do so.
- RQ6: there is a positive correlation between **perceived behavioral autonomy and self-efficacy** to be able to share data and a researcher’s intention to do so.
- RQ7: **researchers’ past data sharing behavior** increases with the intention to share data.
- RQ8: a **researcher’s demography** affects the intention to share data: **discipline, country**, and to some extent also **working sector, position in career and age** all play a role in this respect.”

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Melero and Navarro-Molina

https://doi.org/10.1002/leap.1287

101 Food Science and Technology researchers

Not explicitly TPB, but including related terms or questions

Melero and Navarro-Molina (2020), Figure 8 - Important motivators to publish your data as open data

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So what matters - individual attitudes and characteristics

Attribute	**McEachern & McDougall (2014)**	**Kim and Stanton (2015)**	**Zenk- Moltgen et al. (2018)**	**Melero and Navarro-Melina (2020)**
Attitudes towards data sharing	++	(Scholarly altruism ++)	+	+
Subjective norms	+	++	+	+
Perceived behavioural control	NO	(Effort: -)	+	
Demographics	Female (~)		Discipline, country, career stage, age	
Past data sharing	++		+	
Interest in secondary use of others' data	+++			

So what matters? Institutional characteristics

Attribute	McEachern & McDougall (2014)	Kim and Stanton (2015)	Zenk-Moltgen et al. (2018)	Melero and Navarro-Melina (2020)
Data repository availability		NO		
Institutional arrangements/RDM	NO			
Funder requirements	NO			+
Journal requirements		++		+

So what do we need?

Policies and infrastructure are important for understanding **“supply”** of data sharing capacity

Better understanding of researcher motivations to understand “demand” for data sharing activity

Sticks (funder and journal requirements) matter

But so do disciplinary norms and personal attitudes

So what might influence FAIR? Over to Kheeran...